Precision Grasp Force Regulation for Rigid Gripper Systems: Integrating Computer Vision and Large Language Models

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***Abstract*—The integration of artificial intelligence and robotics has significantly advanced the field of robotic grasping, particularly in achieving precision and efficiency. Grasping, a fundamental human skill, has long posed challenges for robotic systems due to the complexity of sensory feedback and cognitive processing involved. This study focuses on enhancing Grasping Force Control (GFC) by leveraging computer vision and local large language models to improve robotic adaptability and decision-making. Unlike traditional sensor-based systems, computer vision offers a cost-effective, human-like approach to object recognition and manipulation, enabling robots to interpret their surroundings more intuitively. We integrated YOLO v11 and a local large language model, to provide prior assumptions via prompts, guiding the gripper in adjusting its force based on detected objects. This approach emulates human-like decision-making, translating visual information into precise motor commands. Addressing the limitations of open-loop control systems, we employ a closed-loop control system to ensure accurate and adaptive grasping force regulation. Our rigid gripper design, enhanced with advanced computer vision and language models, offers precision and consistency, making it suitable for diverse industrial applications. Key contributions include: (1) a novel closed-loop gripper control method based on computer vision, (2) a prompt-based system for adaptive force adjustment, and (3) the use of rigid 1-DoF gripper to enhance adaptability and precision. This research advances robotic grasping force control and demonstrates the potential of integrating computer vision and large language models to achieve human-like precision and adaptability in robotic systems.**

***Index Terms*—Robotic grasping, computer vision, large language models, force control, closed-loop systems, YOLO v11**

I. Introduction

The integration of artificial intelligence and robotics has ushered in a new era of precision and efficiency, particularly in the domain of robotic grasping. Grasping is a fundamental human skill that has long been a challenge for robotic systems to replicate accurately. Many studies have sought to advance the field of Grasping Force Control (GFC) to achieve accurate robotic control [1], [2]. The complexity of human grasping lies not only in the physical act but also in the intricate interplay of sensory feedback and cognitive processing. As robots increasingly interact with dynamic and unpredictable environments, the need for advanced GFC has become important.

In this context, computer vision emerges as a pivotal technology for enhancing robotic grasping capabilities. Unlike traditional sensor-based systems, computer vision offers a cost-effective and human-like approach to object recognition and manipulation. By mimicking the human visual system, computer vision enables robots to perceive and interpret their surroundings in a more intuitive manner. This technology has been successfully applied in various fields, including autonomous driving, where Tesla's reliance on computer vision has demonstrated its potential for real-time decision-making and control.

Inspired by Tesla's computer vision solutions and aiming to enhance the adaptability and intelligence of robotic grippers, we integrated the YOLO v11 visual detection model into our system. This model generates prior assumptions in the form of prompts, enabling the gripper to adjust its force based on the detected object. This integration facilitates human-like decision-making, where visual information is translated into precise motor commands.

Our research also addresses the limitations of existing open-loop control systems, which, despite their reliability and cost-effectiveness, suffer from limited accuracy and susceptibility to disturbances. In contrast, closed-loop control systems offer the advantage of immediate feedback, enabling precise adjustments and minimizing deviations. This study employs a closed-loop control system to ensure optimal performance in grasping force regulation.

The rigid gripper design chosen for this study is particularly advantageous in applications requiring precision and consistency. While soft grippers have gained attention for their ability to handle delicate objects, rigid grippers remain a staple in many industrial settings due to their reliability and strength. By integrating advanced computer vision algorithms and local large language models, we have developed a novel rigid gripper system that can adapt to various objects and conditions.

Our contributions can be summarized as follows:

- We introduce a new method for controlling closed-loop grippers based on computer vision.
- We develop a prompt-based system that adjusts gripper force according to the detected object.
- We use rigid 1-DoF gripper, designed to enhance adaptability and precision in robotic grasping tasks.

This study not only advances the field of robotic grasping force control but also highlights the potential of integrating computer vision and large language models to achieve human-like precision and adaptability in robotic systems.

II. Related Works

A. Grippers for Grocery Picking

Today, some commercial grippers are assembled on the robotic arms to do picking jobs. Their working method is based on sucking and placing objects. However, such a working method can only work very well on rigid or specific objects. It does not have a good effect on picking soft objects. Therefore, L. Beddow, H. Wurdemann, and D. Kanoulas designed a 3-finger gripper that applied to the grocery-grasping field [3]. They believe that the grocery field faces issues where grocery items may be damaged if the robotic gripper lacks accurate grasping control. To solve this problem, they incorporated force feedback into the gripper, which was trained using a reinforcement learning algorithm.

We can see from the Table I, most low-DoF gripper do not use close loop feedback. Zeng A et al. [4] trained two convolutional neural networks to help the robot learn how to coordinate these two types of actions: one that cannot involve grabbing (like pushing) and the other that involves grabbing (like picking up). Chen Z et al. [5] proposed one framework that trained vision and control separately rather than end-to-end, the pros of this are it can be optimized base on the specific task. Similar with them, our work also separate vision detection and control strategy into two parts. This paper Kim et al. [6] proposed accelerating actor-critic deep reinforcement learning for visual grasping in cluttered environments by using state representation learning that disentangles raw input images, improving efficiency and performance in grasping tasks. Liu et al. [7] proposed a new way to grasp objects using a soft gripper and a smart learning method. This combination helps robots handle complex objects and environments better. Following this, the study in [8] introduces a smart learning system for robots to push and pick objects in messy spaces. It uses active exploration and a special gripper design to improve success rates in tough situations. Next, the work in [9] offers a smart learning system for robot tasks. It uses virtual environments and step-by-step learning to make robots work better in real life. Similarly, the research in [10] presents MAT, a smart learning method for multi-fingered grasping. MAT uses touch and position feedback to improve success rates and handle errors, even when trained only in simulations. In another approach, [11] suggest a cheap way to collect human actions to train a strong grasping model. This helps robots grasp new objects in real-world messy environments. Building on this, the framework in [12] introduces a smart learning system for multi-fingered grasping. It combines vision and touch feedback to learn how to grasp different objects in various settings. Finally, the method in [13] proposes a smart learning system for multi-fingered grasping. It uses touch, joint forces, and position data to learn how to move fingers and lift objects. The training happens in simulations, making it easier to adapt and perform well in real-world tasks.

In summary, these studies collectively demonstrate the importance of integrating advanced learning algorithms, feedback mechanisms, and multimodal data to improve robotic grasping capabilities across diverse applications and environments.

B. FOC Close Loop Control

Motor control is an important part of the automation control field. Different speed control methods have significant effects in the motor control industry. As a smooth and steady speed control method, field-oriented current control (FOC) has the advantage of precise control actions compared to direct torque control [14]. It divides torque and magnetizing flux into two parts and weakens the association between them. The speed of the motor is adjusted by controlling the torque directly while magnetization remains unchanged during the low-speed phase.The basic logic of the FOC closed-loop control process is demonstrated in Figs. 1 and 2. A comparative diagram between open-loop and closed-loop control systems is shown in Fig. 3.

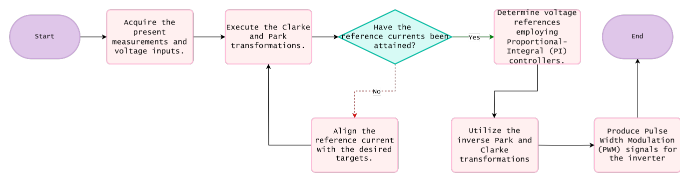

Fig. 1: FOC close loop control process.

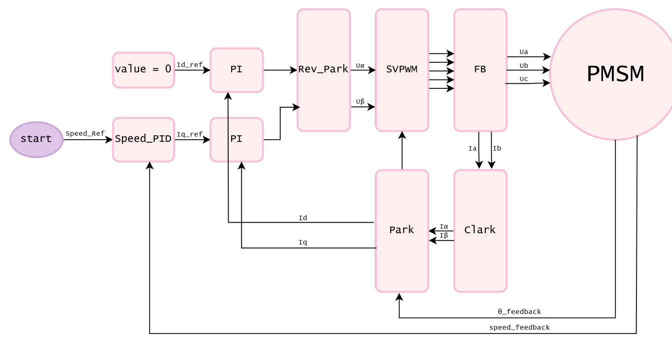

Fig. 2: Schematic representation of the Field-Oriented Control (FOC) mechanism.

Explanation of Parameters

The control process is initiated at the **Start** point. The parameter `Speed_PID` refers to a **Proportional-Integral-Derivative (PID)** controller, which is pivotal for speed regulation. It dynamically adjusts the motor speed to align with a predefined setpoint. The parameter `Value = 0` signifies an initialization step, where a parameter or variable is reset to its baseline state, which is zero in this context.

The **PI controller** (Proportional-Integral controller) is employed within the current loop to regulate the d-axis and q-axis currents. This control action is essential for managing both the magnetic field and torque. `Rev_Park` denotes the inverse Park transformation, which translates the current from the rotating dq reference frame back to the stationary three-phase (ABC) reference frame. Conversely, the **Park transformation** facilitates the conversion of three-phase AC motor currents or voltages from the stationary ABC reference frame to the rotating dq reference frame, thereby enabling effective control within the dq frame.

SVPWM (Space Vector Pulse Width Modulation) is a sophisticated modulation technique used in inverters to regulate the switching states and generate the requisite three-phase AC voltage for motor propulsion. The **Clarke transformation** is

TABLE I: Comparison between different gripper algorithms of grasping

Paper	Gripper Type	DoF	Close Loop Feedback Using	Integrated AI Algorithm
Learning Synergies between Pushing and Grasping with Self-supervised Deep Reinforcement Learning [4]	Two finger gripper	1	No	Two fully convolutional networks
Towards Generalization and Data Efficient Learning of Deep Robotic Grasping [5]	Two finger gripper	1	No	Deep reinforcement learning
Acceleration of Actor-Critic Deep Reinforcement Learning for Visual Grasping by State Representation Learning Based on a Preprocessed Input Image [6]	Two finger gripper	1	No	Deep reinforcement learning
Our work	Two finger gripper	1	Yes	Deep learning
Hybrid Robotic Grasping With a Soft Multimodal Gripper and a Deep Multistage Learning Scheme [7]	Four finger gripper	2	No	Deep reinforcement learning
Deep Reinforcement Learning for Robotic Pushing and Picking in Cluttered Environment [8]	Two finger gripper	3	No	Deep reinforcement learning
Pixel-Attentive Policy Gradient for Multi-Fingered Grasping in Cluttered Scenes [9]	Three finger gripper	4	No	Reinforcement Learning
MAT: Multi-Fingered Adaptive Tactile Grasping via Deep Reinforcement Learning [10]	Three finger gripper	4	Yes	Deep reinforcement learning
Grasping in the Wild: Learning 6DoF Closed-Loop Grasping from Low-Cost Demonstrations [11]	Two finger gripper	6	Yes	Deep reinforcement learning
Contextual Reinforcement Learning of Visuo-tactile Multi-fingered Grasping Policies [12]	Four finger gripper	16	Yes	Deep Reinforcement learning
Multifingered Grasping Based on Multimodal Reinforcement Learning [13]	Five finger gripper	20	Yes	Multimodal Reinforcement Learning

the initial step in current control for FOC, converting three-phase AC motor currents or voltages from the three-phase stationary reference frame to the two-phase stationary ($\alpha\beta$) reference frame.

FB (Feedback) is the process that compares the actual motor state—such as speed and current—with the desired state, thereby informing adjustments to the control strategy.

The **PMSM** (Permanent Magnet Synchronous Motor) is the device under the purview of the FOC strategy, which aims to achieve efficient and precise motor control.

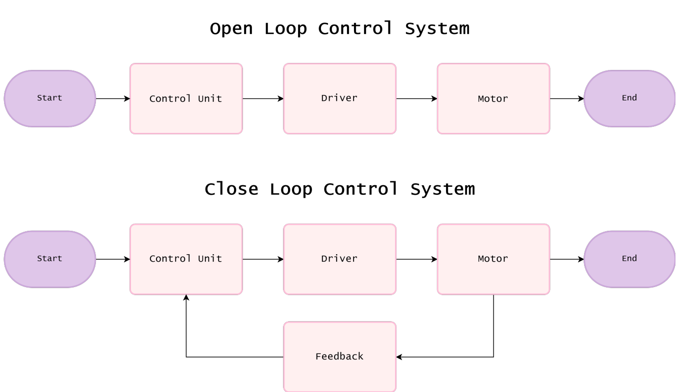

Fig. 3: Comparative analysis of open-loop and closed-loop systems.

C. Computer Vision Algorithm for Grasping

Compared to using 3D sensing cameras and sensors, computer vision costs less because it uses fewer hardware devices. Besides the low-cost advantage, computer vision has biomimetic advantages. This means that computer vision technology, to some extent, mimics the human visual system, making it more intuitive and natural in processing visual information. The giant in the electric car industry, Tesla, uses only computer vision to process data and neural networks to enable the computer to understand how it represents. It has already gained an advanced position in the autonomous driving field, which also proves the accessibility of using computer vision. Transferring this point from autonomous driving to the accurate grasping of robotic grippers, we know that computer vision can be a useful tool to improve performance in the grasping force control field, so we integrate a mature computer vision framework into our project. Computer vision grasping mainly consists of three parts: robotic arm control, vision detection, and robotic gripper control [15]. We will only focus on the last two parts in this article (vision detection and robotic gripper control). These two parts will be more closely associated with each other in our design architecture. They will function as one integral component and will interact with each other. For vision detection, it is distinguished by two different paths: the analytical path and the data-driven path [16]. The analytical path can also be called an experience-based method, as it is based on human prior knowledge and specific environments for designed problems [17]. It originally stems from the experience gained through daily practice since humans were born Therefore, people will make a judgment about the target object before they prepare to grasp it. These analysis steps are usually processed by the brain subconsciously and are imperceptible to people. It is applied to different application scenarios. The first one will only detect familiar objects using the detection model, which is like a CNN image classification model in machine learning, where the model, lacking generalization capabilities, can only identify specific images. In contrast, the data-driven approach has generalization ability, enabling it to detect and grasp unknown objects based on previous training of known object detection. It can be explained as an overfitting-solved method that not only succeeds in grasping seen objects or data based on the training phase. Limited by our type of computer vision algorithm, this research adopted analytical methods, which means the robotic gripper operates under prior knowledge assumptions and grasps specific objects selected by researchers.

D. CANController Area Networkapplication in Robotics field

Research paper [18] proposed a decentralized framework based on the CAN bus, which identifies the advantages of CAN compared to traditional robot hardware architecture: decentralization, instant communication, reliability, and flexibility. Firstly, for decentralization, this type of architecture reduces reliance on the central processing unit and increases the fault tolerance of the system. Secondly, for instant communication, it is crucial for the robot to respond to changes in the surrounding environment. Thirdly, the design of the CAN bus is reliable, allowing the robot to function even in electrically noisy environments. Fourthly, it is easy to add and remove endpoints, making the system design more flexible. However, there are some drawbacks: it requires additional software and hardware deployment costs, and decentralization also increases the complexity of system design. From an application perspective, research paper [19] discussed how to control a 6-DOF robotic arm by using a CAN interface to communicate with ROS (Robot Operating System). This 6-DOF robotic arm also proves that adopting the CAN bus for communication is efficient and reliable, making it well-suited for instant control application scenarios. The future direction of work is to optimize the control algorithm to improve the responsiveness and extensibility of robotic systems.

E. Large language model in robotic manipulation

Artificial Intelligence has increasingly strong learning capabilities due to the rapid development of artificial neural networks [20].As one the most important scope of AI, LLM has dominated the mainstream position in natural language process. Not only can LLM encode rich semantic knowledge relate to the world, but also can apply the prior knowledge to improve robots high-level action performance [21]. Nowadays, many studies have leveraged LLMs to enhance robots' adaptability in complex environments. They achieve this by using prompts that provide available strategies or coding to guide the robots' actions and decision-making processes [22]–[31]. The most representative application of using LLMs for robotic manipulation is the CoPa framework [32]. CoPa is a novel framework that uses a Visual Language Model (VLM) to help robots move and handle objects. It generates a series of positions for the robotic arm. This system leverages the basic knowledge that the VLM has about how the physical world works, which is similar to how people intuitively understand everyday things without much conscious effort. Similar to CoPa, we also designed an interface as a bridge between LLM and robot manipulation. Their interface is based on spatial geometry constraints, whereas our interface is based on the maximum output voltage of the motor in closed-loop mode.

III. Gripper Design

A. Hardware Design

Our hardware architecture centers on a single stepper motor as the primary actuator, integrated with 3D-printed components. The screw parameters include a diameter of 5 mm, a lead of 2 mm, and a length of 34 cm. The control board is based on the ZDT Emm28_V5.0, which employs a Cortex-M4F core operating at a frequency of 120 MHz. For electrical communication, we have implemented the ZDT_CAN module for CAN Bus communication. We utilize the USBCAN CANable adapter for linking the gripper's control board with a laptop. This adapter supports baud rates up to 1M and features two pins, compatibility with USB2.0 and TYPE-C interfaces, and a load resistance of 120 . Power is supplied by a 24 V, 3 A adapter from ZDT. The hardware architecture as shown in Fig. 4.

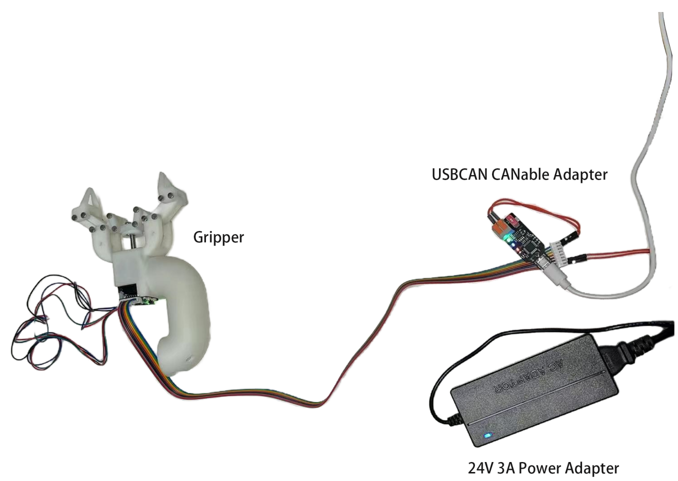

Fig. 4: Hardware Architecture.

B. Software Design

In terms of software architecture, we have adopted Python as the programming language for driver development. This choice ensures consistency and allows for seamless integration with object detection algorithms. We use PEAK Drivers 4.5.0 from PEAK-System for CAN bus communication. Our design philosophy follows a modular construction approach, offering flexibility to interchange components based on specific needs, adhering to the software engineering principle of high cohesion and low coupling. We employ YOLO and a local LLM llama3.2 as the main driving software instead of a VLM (Vision Language Model). From an efficiency standpoint, YOLO outperforms VLM in terms of rapid object detection response times, ensuring a smooth operational flow without interruptions.Our work ensures safety by operating without requiring an internet connection. This reduces the risk of data loss and protects against hacking. Economically, YOLO is more resource-efficient and can be effectively deployed on non-GPU computer instances. We achieve comparable results by combining YOLO with a local LLM instead of utilizing a VLM. This is important because many advanced systems, despite their superior performance, are too expensive to be practical in real-world situations.

We have developed prompts to enhance the capabilities of LLMs by emulating the function of tactile sensors. These prompts leverage existing human understanding and assessments of the tactile characteristics of objects, minimizing the need for physical sensors. This approach aligns with the human brain's operational mechanism, which relies on pattern recognition and associative learning. Our design prioritizes the brain's role over tactile feedback, reflecting the human tendency to preset grip force before grasping—a mechanism originating from the brain and derived from tactile training and knowledge memory accumulation. These prompts function as databases of learned experiences and heuristics, similar to neural pathways that link sensory inputs with specific responses in the brain. By integrating these prompts, our system can anticipate tactile feedback without physical contact, similar to how humans predict object texture or touch outcomes based on memory.

This design not only reduces hardware costs but also enhances the system's adaptability. It learns from each interaction and refines its knowledge base, similar to the human brain. Shifting complexity from hardware to software offers a cost-effective and scalable solution that can be upgraded through software updates, eliminating the need for costly hardware fixes.

This study harnesses the capabilities of a large language model (LLM) to translate natural language prompts into executable robotic code [21]. The LLM is trained with real-world constraints, instructing the robot not to directly enact the prompt's intention but to perform the most likely successful task outcome. Extensive empirical research has validated the feasibility of using LLMs to generate code for robotic manipulation tasks [33]–[38]. This approach is termed "code policy" [39], where the generated code invokes primitive application programming interface (API) methods. To illustrate this concept, consider a hypothetical control function designed to accept the name of a target object and directional parameters x and y. This function updates the object's x-pos and y-pos attributes and outputs the new coordinates. In the context of code policy, the LLM generates the required parameters x and y and subsequently inputs them into the move object algorithm. As shown in Algorithm 1, this algorithm is a simple example that demonstrates how the object's position is updated based on the input parameters.

Algorithm 1 Move Object Algorithm

Input: item, x, y
Output: None
function MOVEOBJECT(item, x, y)
  item.x_pos $\leftarrow$ item.x_pos $+ x$
  item.y_pos $\leftarrow$ item.y_pos $+ y$
  `print("Moved object to", item.x_pos, item.y_pos)`
end function

Consequently, the LLM can convert high-level, natural language directives into specific, actionable low-level control commands, enabling precise manipulation of robotic systems. The primary advantage of this methodology is its utilization of the sophisticated language comprehension skills of pre-trained LLMs, thereby reducing reliance on manual coding and enhancing the system's adaptability and generalization to novel instructions. Drawing inspiration from the code policy paradigm, we have designed our software workflow, as shown in Fig. 5.

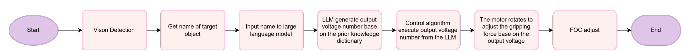

Fig. 5: Software workflow.

IV. EXPERIMENT

Firstly, we only test in a simulation environment. The successful accurate grasping criteria consist of only one item: (a) The difference between the output torque of the algorithm and the previously predicted torque; the smaller the difference, the higher the grasping accuracy. The previously predicted torque for different objects is derived from practical testing. The picture below shows the relationship between the maximum motor input voltage and the maximum output torque. We control the output torque by adjusting the input voltage. A simple calculation transformation process between maximum output torque and maximum output voltage is shown in Fig. 6.

Secondly, we tested in the real world using a real gripper to conduct the experiment. The successful accurate grasping criteria were: (a) The object can be held successfully. (b) The object was not affected or interrupted during grasping. (c) The difference between the output torque of the algorithm and the previously predicted torque; the smaller the difference, the higher the grasping accuracy. We assembled the gripper with the control unit and ran the AI algorithm to conduct the experiment. We did not conduct a comparison experiment; instead, we are trying to use different types of grippers to achieve the best intelligent effect in accurate grasping control. What we did is attempt to enable the computer to understand the object and automatically adjust the grasping force to an appropriate value by adjusting the maximum output voltage of the motor. In the real world, we tested 12 objects and verified that it is feasible to improve the accurate control of robotic gripper grasping. The objects tested span a variety of categories, including everyday utensils such as bottles, cups, and

To convert the maximum output voltage from millivolts (mV) to volts (V), we use the following conversion:

$$V = \frac{5000 \text{ mV}}{1000} = 5 \text{ V}$$

Now, we can use this voltage and the given rated current to calculate the power:

$$P = V \times I = 5 \text{ V} \times 0.6 \text{ A} = 3 \text{ W}$$

To convert the power from watts to kilowatts:

$$P_{kW} = \frac{3 \text{ W}}{1000} = 0.003 \text{ kW}$$

Using the power and rotational speed, we can calculate the torque:

$$T = \frac{9550 \times P_{kW}}{n} = \frac{9550 \times 0.003}{500} = \frac{28.65}{500} = 0.0573 \text{ Nm}$$

To convert the torque from Newton meters to Newton centimeters (since 1 Nm = 100 Ncm):

$$T = 0.0573 \text{ Nm} \times 100 = 5.73 \text{ Ncm}$$

So, when the maximum output voltage is 5000 mV (or 5 V), the torque is approximately 5.73 Ncm.

Fig. 6: Relationship between maximum output torque and maximum output voltage.

spoons; fruits like bananas, apples, and oranges; vegetables such as broccoli and carrots; and common household items like remotes, cell phones, books, scissors, and toothbrushes. These objects were categorized into different layers (Layer A, Layer B, Layer C, Layer D, and Layer E) based on their properties and the complexity of grasping them.

Finally, the algorithm was executed to grasp the 12 objects, and the results were observed. The grasp experiments were recorded as videos, which are available on our research project's GitHub showcase page: https://2441630833.github.io/foc-gripper-grasping/. The grasp effectiveness was evaluated based on the previously established criteria for successful and accurate grasping.

V. Results

The result of the grasping experiment in a real environment is shown by Table II. The system achieved 100% grasp success across 12 objects with varying geometries and material properties, demonstrating robust closed-loop control. A single voltage bias (8.33%) occurred during apple grasping, likely due to its smooth, deformable surface challenging the vision-to-force mapping. However, real-time feedback mechanisms compensated for this error, ensuring a stable grasp. The torque range (1.146–5.73 Ncm) correlated with object-specific requirements, validated by precise voltage alignment (0% bias for 11/12 objects). These results highlight the synergy of computer vision and adaptive control in enabling reliable robotic manipulation for diverse real-world tasks.

TABLE II: Grasp Result

Objects	Hold	Not Affected	Voltage (mV)		Bias
			Exp.	Act.	
bottle	Yes	Yes	4000	4000	0
cup	Yes	Yes	1000	1000	0
spoon	Yes	Yes	5000	5000	0
apple	Yes	Yes	3000	2000	1000
orange	Yes	Yes	2000	2000	0
broccoli	Yes	Yes	3000	3000	0
carrot	Yes	Yes	3000	3000	0
remote	Yes	Yes	4000	4000	0
cell phone	Yes	Yes	4000	4000	0
book	Yes	Yes	4000	4000	0
scissors	Yes	Yes	5000	5000	0
toothbrush	Yes	Yes	5000	5000	0

VI. Discussion & Limitations

Although our gripper performs well in accurate force grasping, it has some limitations. Firstly, it belongs to transient learning rather than long-term persistent learning because our prompt is assumed to be already known by the LLM. It only learns once and does not receive updates after the initial training. Secondly, compared to reinforcement learning, the code policy strategy is more efficient for quick application in real-world scenarios. It does not require a long training process but only a prior assumption (which comes from the author of the code). In this work, we solve the grasping control problem by incorporating a prior assumption (creating two prompts and setting a constraint prompt for the LLM). The LLM does not understand what is happening in the gripper and only addresses one part of the problem. We only utilize the logic processing part of the LLM, similar to a factory pipeline. This mode allows for quick deployment in different scenarios but lacks strong generalization ability. Thirdly, there must be a trade-off. It is impossible to maintain both strong generalization ability and high accuracy simultaneously. In our work, we sacrifice generalization to maintain a high success rate in accurate grasping control. This limitation means that our gripper is designed to perform specific tasks rather than handle a wide range of scenarios. While it excels in precise and controlled environments, it is less suited for applications requiring broad adaptability. As a result, our approach is more specialized, focusing on achieving high performance in targeted use cases rather than general-purpose functionality. Lastly, rigid grippers offer higher response speeds but are more susceptible to damage. In contrast, soft grippers can mitigate mechanical fatigue through their structural design. Which one is better overall? Even more promising are bionic robot grippers. The study by Zhanchi Wang, Nikolaos M. Freris, and Xi Wei [40] presents a new type of soft robot called SpiRobs, which mimics the logarithmic spiral pattern found in natural structures like octopus arms and elephant trunks. Unlike traditional soft grippers, SpiRobs demonstrate better adaptability and scalability. This is evidenced by their successful use in different forms, such as a small gripper, a 1-meter-long manipulator attached to a drone, and a group of SpiRobs that can entangle multiple objects. This bionic design, inspired by nature, provides a strong alternative to both rigid and conventional soft grippers, combining flexibility, durability, and adaptability for robotic grasping tasks. Looking ahead, such grippers hold immense potential for broader applications, paving the way for more advanced and versatile robotic systems in the future.

VII. Conclusion

In this study, we developed a robotic grasping system that combines computer vision and local large language models

(LLMs) to achieve precise and adaptive force control. By using YOLO v11 and prompt-based decision-making, our system mimics human-like reasoning to adjust grasping force based on object detection. The closed-loop control mechanism ensures high accuracy, achieving a 100% success rate in grasping experiments across 12 objects with varying shapes and materials. However, our approach has limitations, such as transient learning and a trade-off between generalization and accuracy, making it more suitable for specific tasks rather than broad applications. For future work, the development of bionic designs like SpiRobs will be considered.

ACKNOWLEDGMENT

The authors would like to thank the anonymous reviewers for their valuable comments and suggestions.

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